HESOD: DUAL-EVIDENCE SELECTION AND COVERAGE OPTIMIZATION FOR HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION

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ABSTRACT

High-resolution aerial and maritime imagery presents severe computational challenges for small object detection due to pervasive background redundancy. While selective-computation detectors dynamically route processing to candidate regions, existing selectors rely strictly on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse, causing irreversible early pruning under an asymmetric selection cost. In this paper, we propose HESOD, an efficient dual-evidence selection framework tailored for recall-safe small object detection. HESOD introduces a dual-evidence priority head that fuses semantic objectness with channel-pooled spectral saliency at negligible $\mathcal{O}(1)$ compute overhead, recovering weak semantic targets. Furthermore, we formulate an object-level soft-coverage loss that directly maximizes joint instance coverage probability, fundamentally preventing premature false negatives. Paired with a lightweight inverted residual decoupled head and scale-aware Wasserstein regression, HESOD achieves +2.7 pp mAP and +5.5 pp total recall gains on UAVDT, and improves mAP by +1.9 pp on SeaPerson with 27.6% fewer parameters while sustaining real-time throughput at 77.1 FPS.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, search-and-rescue, and environmental monitoring [1, 2]. In these applications, targets of interest typically occupy merely a handful of pixels ($< 16 \times 16$ px), while uniform background dominates over 70% to 80% of the spatial area [3, 4]. Executing dense deep detectors across multi-megapixel grids incurs prohibitive arithmetic and memory overhead on edge platforms.

To eliminate redundant computations, *spatial selective computation* has emerged as an effective paradigm [3, 5–7]. Frameworks such as QueryDet [6] and CEASC [7] sparsify feature pyramids, while ESOD [3] predicts an early-stage spatial heatmap after the stem to route only candidate foreground patches to downstream stages.

Despite substantial compute savings, existing spatial selectors suffer from a critical vulnerability: they rely exclusively on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse into background noise [8]. More crucially, spatial routing decisions exhibit an *asymmetric cost*: retaining an extra background patch only marginally increases latency, whereas discarding a patch with a true tiny target creates an irrecoverable false negative for all subsequent stages [3]. Furthermore, standard pixel-wise cross-entropy supervision evaluates grid cells independently, providing no guarantee that at least one candidate cell covers a given ground-truth instance.

To overcome these limitations, we observe that tiny targets preserve distinctive high-frequency structural gradients and spectral saliency even when high-level semantic activations fade [8]. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a framework integrating multi-evidence routing with coverage-constrained spatial optimization. Specifically, HESOD introduces an ultra-lightweight dual-evidence priority head that fuses semantic objectness with channel-pooled spectral saliency at strictly $\mathcal{O}(1)$ complexity relative to backbone width. To ensure recall safety, we design an object-level soft-coverage loss that directly maximizes the joint coverage probability of ground-truth instances. Finally, an inverted residual partial-convolution (ISPP) decoupled head and scale-aware balancing loss (SABL) [9] are integrated to accelerate inference and stabilize micro-scale box regression.

Our key contributions are summarized as follows:

1. We formulate an object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) that directly maximizes joint instance coverage probability, fundamentally preventing premature false negatives in selective computation.
2. We design a dual-evidence priority head combining high-level semantic objectness with channel-pooled

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spectral saliency at $\mathcal{O}(1)$ complexity relative to channel width, rescuing targets with collapsed semantic activations.

3. We develop a lightweight deployment engine integrating an inverted residual decoupled head (ISPP) and scale-aware Wasserstein regression, achieving superior recall–accuracy–efficiency trade-offs with 27.6% fewer parameters.

2. RELATED WORK

2.1. Small Object Detection in Aerial Imagery

Detecting micro targets in aerial and maritime imagery is severely hindered by extreme scale variations, low contrast, and complex backgrounds [1, 2, 4, 10]. Classical methods employ image cropping or density-guided patch generation (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]), which often require separate coarse detector networks or incur repeated feature extraction on overlapping tiles. Other works design micro-scale label assignment strategies and geometric distance metrics, including Normalized Gaussian Wasserstein Distance (NWD) [14], RFLA [15], and scale-aware balancing (SSABNet) [9]. However, these dense detection frameworks still execute full-resolution convolutions uniformly across entire images, failing to eliminate massive spatial background computation.

2.2. Selective and Dynamic Computation

To reduce arithmetic redundancy, selective computation dynamically routes processing to prospective foreground regions. AutoFocus [5] predicts focus pixels to guide multi-scale processing. QueryDet [6] accelerates feature pyramids via cascaded sparse queries, while CEASC [7] formulates adaptive activation masking for sparse convolutions. Recently, ESOD [3] introduced an early ObjSeeker module after the shallow stem to slice features into patches (AdaSlicer), bypassing background across subsequent stages. Nonetheless, existing selectors rely strictly on a single semantic objectness score and independent per-cell BCE loss, leaving them vulnerable to semantic feature collapse and irrecoverable tiny-target drops.

2.3. Spectral Saliency and Frequency Methods

Frequency-domain analysis provides complementary spatial representations. SET [8] demonstrated that tiny objects preserve distinct high-frequency structural signatures even when semantic activations fade, using frequency background smoothing to enhance dense feature maps. Unlike SET, which applies dense spectral operations over entire multi-scale pyramids, HESOD repurposes spectral and gradient saliency as an ultra-lightweight, channel-pooled routing signal ($\mathcal{O}(1)$ relative to backbone width) to guide recall-safe spatial selection without dense frequency overhead.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 1(a). Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a shallow stem extracts dense features $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$. The *Dual-Evidence Spatial Selector* then operates in two coordinated stages: (i) a Dual-Evidence Priority Head fuses semantic objectness with channel-pooled spectral saliency to generate a continuous priority map $\mathbf{S} \in [0,1]^{H_s\times W_s}$; and (ii) an AdaSlicer routing module dynamically crops high-priority candidate feature patches $\mathcal{P}$ directly from $\mathbf{F}$. Downstream backbone, neck, and lightweight ISPP decoupled detection heads evaluate only these retained patches, trained end-to-end via an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

Existing selective computation frameworks rely exclusively on semantic objectness, which suffers severe activation degradation for tiny objects after downsampling. To recover weak semantic targets, HESOD introduces a dual-evidence priority mechanism combining high-level semantic confidence with low-level structural saliency at negligible compute overhead (Fig. 1(b)).

Let $f_i \in \mathbb{R}^C$ denote the feature at spatial location i of shallow feature $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $z^{\text{sem}}_{i,c} = \text{Conv}_{1\times 1}(f_i), c \in \{0,\dots,N_c-1\}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C}\sum_{c=1}^{C}\mathbf{F}_{:,:,c}\right], \tag{1}$$

where $[\cdot \,\|\, \cdot]$ denotes concatenation, yielding $\mathbf{F}_{\text{pooled}} \in \mathbb{R}^{H_s\times W_s\times 2}$. Compressing $\mathbf{F}$ to 2 channels guarantees that subsequent spatial filtering complexity remains strictly $\mathcal{O}(1)$ regardless of backbone width C.

Next, a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian and directional Sobel operators—is applied over $\mathbf{F}_{\text{pooled}}$ to capture isotropic edge transitions and directional gradients. A 1×1 projection layer maps the 6 filtered channels back to dimension C to output spectral logits $z^{\text{spec}}_{i,c}$. Finally, semantic and spectral logits are concatenated and fused via a 1×1 convolution to predict class-agnostic routing priority scores $s_i = \sigma(u_{i,0}) \in [0,1]$:

$$u_{i,c} = \text{Conv}_{1\times 1}\left(\left[z^{\text{sem}}_{i,c} \,\|\, z^{\text{spec}}_{i,c}\right]\right). \tag{2}$$

Candidate regions satisfying $s_i > \tau$ are clustered into bounding patches and routed to downstream stages.

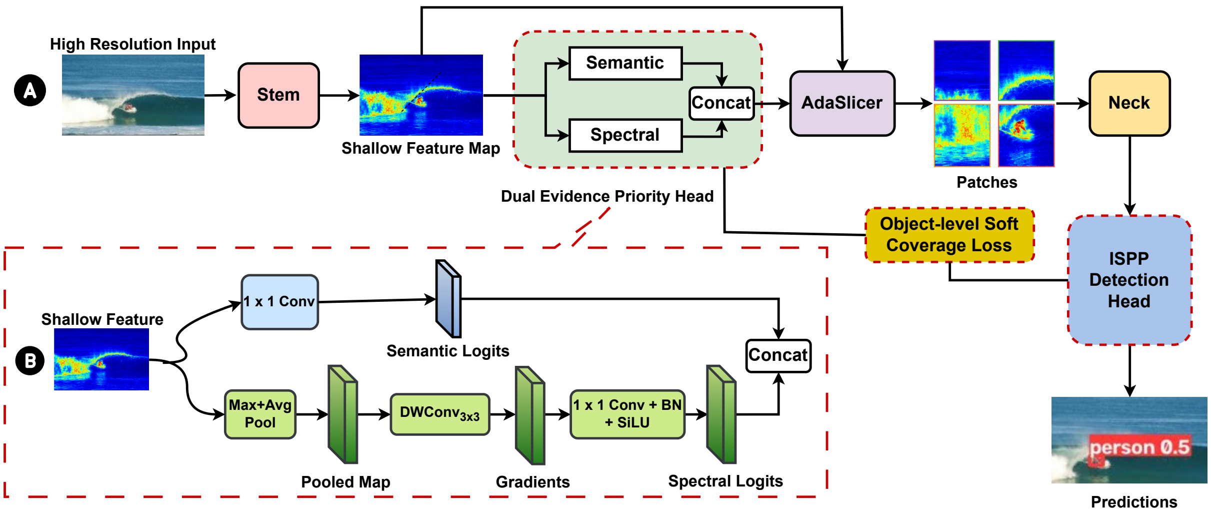

Fig. 1. Overview of HESOD: (a) End-to-end selective detection pipeline comprising the Stem, Dual-Evidence Priority Head (with $\mathcal{L}_{\text{cover}}$), AdaSlicer router, Neck, and ISPP Head; (b) Inner structure of the Dual-Evidence Priority Head. Red dashed boxes highlight our proposed architectural components and loss enhancements over baseline ESOD.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance coverage: a selector may over-allocate cells to large objects while dropping tiny objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. Assuming independent cell activations, the probability that instance j is covered by at least one selected cell is:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{3}$$

The soft-coverage loss directly optimizes joint instance coverage:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log\left(p_j^{\text{cover}} + \epsilon\right), \tag{4}$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in grid cells, penalizing misses on micro targets ($a_j < 4$). The total training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Efficient Decoupled Head and Scale-Aware Loss

To reduce computational overhead while boosting localization precision, we incorporate an inverted residual partial-convolution (ISPP) decoupled head [16] and Scale-Aware Balancing Loss (SABL) [9].

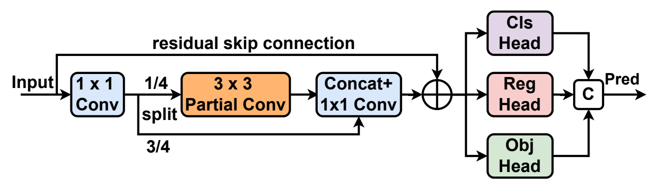

Fig. 2. Architecture of the Inverted Residual Decoupled Head (ISPPHead) with Partial Convolution (Ⓒ denotes channel concatenation).

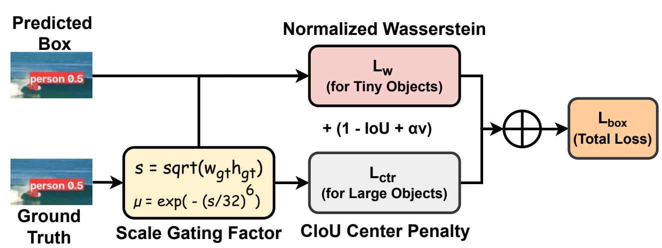

Fig. 3. Scale-Aware Balancing Loss (SABL) workflow.

As shown in Fig. 2, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, drastically lowering compute while eliminating task interference.

Furthermore, to mitigate gradient vanishing caused by IoU sensitivity on micro targets, SABL dynamically balances Gaussian Wasserstein distance (D_W) and Complete

Table 1. Detection results on UAVDT (1280×1280).

Method	mAP @.5	mAP @.5:.95	Params (M)	GFLOPs	FPS
Faster R-CNN [17]	*[TBD]*	*[TBD]*	41.5	*[TBD]*	*[TBD]*
RetinaNet [18]	*[TBD]*	*[TBD]*	34.0	*[TBD]*	*[TBD]*
ESOD [3]	<u>0.344</u>	<u>0.171</u>	<u>35.8</u>	*[TBD]*	*[TBD]*
HESOD (Ours)	**0.371**	**0.195**	**25.9**	*[TBD]*	*[TBD]*
Delta (Δ)	**+2.7 pp**	**+2.4 pp**	**-27.6%**	*[TBD]*	*[TBD]*

Bold: best; <u>underline</u>: second-best.

Table 2. Detection results on SeaPerson (2048×2048).

Method	mAP @.5	mAP @.5:.95	Params (M)	GFLOPs	FPS
Faster R-CNN [17]	0.551	0.246	43.3	1546.8	23.1
RetinaNet [18]	*[TBD]*	*[TBD]*	34.0	*[TBD]*	*[TBD]*
ESOD [3]	<u>0.750</u>	<u>0.320</u>	<u>35.8</u>	**202.4**	**85.7**
HESOD (Ours)	**0.769**	**0.323**	**25.9**	<u>209.1</u>	<u>77.1</u>
Delta (Δ)	**+1.9 pp**	**+0.3 pp**	**-27.6%**	+6.7	-8.6

Bold: best; <u>underline</u>: second-best.

IoU (CIoU) via scale gating $\mu(s) = \exp(-(s/32)^6)$ (Fig. 3):

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\Big(1 - e^{-D_W/12}\Big) + (1-\mu(s))\ell_{\text{ctr}} + \alpha v. \tag{5}$$

When target scale $s \to 0$, $\mu(s) \to 1$, ensuring continuous distance gradients even under zero spatial IoU overlap.

4. EXPERIMENTS

4.1. Experiment Settings

We conduct experiments on two challenging high-resolution benchmarks tailored for small object detection under massive background redundancy: **UAVDT** [2] (1280×1280 pixels) and **SeaPerson (TinyPersonV2)** [4, 10] (2048×2048 pixels). UAVDT evaluates urban aerial vehicle detection across three classes (*car*, *truck*, *bus*) under varied altitudes and camera angles. SeaPerson evaluates maritime search-and-rescue over 300,375 annotated instances across 5,752 whole frames, where over 85% of targets are tiny/micro objects ($< 32 \times 32$ pixels). These two datasets satisfy our target scenarios of high resolution, severe scale variations, and sparse foreground distribution without artificial pre-cropping.

All models are implemented in PyTorch using YOLOv5m as the base detector for fair comparison. Training is conducted on an NVIDIA GPU using SGD with cosine annealing for 50 epochs, following standard selective detection protocols [3]. Detection accuracy is assessed using COCO-style $\mathrm{mAP}@.5$ and $\mathrm{mAP}@.5:.95$, alongside size-stratified detector recall across four physical pixel bins: *Very Tiny* ($< 16^2$ px), *Tiny* ($16^2 - 32^2$ px), *Small* ($32^2 - 96^2$ px), and *Med/Large* ($> 96^2$ px). Computational efficiency is evaluated via parameter count (Params), total GFLOPs, frame rate (FPS at batch size 1), and Bounding Patch Recall (BPR).

4.2. Main Benchmark Comparisons

Tables 1 and 2 evaluate HESOD against representative dense detectors and the state-of-the-art selective detector ESOD. On the urban aerial UAVDT benchmark (1280×1280), HESOD outperforms ESOD by **+2.7 pp** in $\mathrm{mAP}@.5$ (0.344 vs. 0.371) and **+2.4 pp** in $\mathrm{mAP}@.5:.95$, with significant per-class vehicle recall improvements (*car* recall rises by **+5.69 pp** from 84.87% to 90.56%). On ultra-high-resolution maritime SeaPerson (2048×2048), HESOD improves $\mathrm{mAP}@.5$ by **+1.9 pp** ($0.750 \to 0.769$) over ESOD while compressing deployed parameters by **27.6%** ($35.8\text{M} \to 25.9\text{M}$) and sustaining real-time inference at 77.1 FPS. Compared to the classical dense detector Faster R-CNN (0.551 $\mathrm{mAP}@.5$, 1546.8 GFLOPs, 23.1 FPS), HESOD achieves a **+21.8 pp** accuracy gain with a **7.4**$\times$ reduction in GFLOPs and a **3.3**$\times$ speedup, confirming the vital necessity of selective computation for high-resolution edge vision.

Table 3. Module ablation on UAVDT (1280×1280).

Method / Config	mAP @.5	mAP @.5:.95	BPR	GFLOPs	FPS
(1) Baseline (BCE)	0.344	0.171	*[TBD]*	*[TBD]*	*[TBD]*
(2) Semantic-only*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(3) Spectral-only*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(4) Spectral-Pooled*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(5) Dual-Concat	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(6) Dual+SABL	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(7) Dual+ISPP	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(8) HESOD (Full)	**0.371**	**0.195**	*[TBD]*	*[TBD]*	*[TBD]*

*Trained with soft-coverage loss $\mathcal{L}_{\text{cover}}$. **Bold**: best.

4.3. Ablation Studies and Analysis

Effectiveness of Dual Evidence and Coverage Optimization. Tables 3 and 4 dissect the component-wise contributions across benchmarks. On SeaPerson (2048×2048), replacing standard BCE with the object-level soft-coverage loss $\mathcal{L}_{\text{cover}}$ (Arm $1 \to 2$) delivers an immediate **+1.9 pp** boost in $\mathrm{mAP}@.5$ and raises Bounding Patch Recall (BPR) from 0.947 to **0.991**, demonstrating that joint instance coverage optimization effectively prevents premature pruning of micro targets. Furthermore, comparing full-width unpooled spectral routing (Arm 3, 0.770 $\mathrm{mAP}@.5$) with its 2-channel pooled counterpart (Arm 4, 0.767 $\mathrm{mAP}@.5$, 0.988 BPR) confirms that our $\mathcal{O}(1)$ channel pooling retains near-identical routing capability without full-width memory overhead. Dual-evidence feature concatenation (Arm 5) achieves peak $\mathrm{mAP}@.5$ (**0.772**) and recovers +420 to +631 more micro targets ($< 16^2$ px) than single-evidence selectors.

Efficiency and Localization with ISPP and SABL. Replacing the conventional detection head with the inverted residual ISPP decoupled head (Arm 7) cleanly reduces arithmetic complexity by **22.6%** ($281.2 \to 217.6$ GFLOPs) and cuts parameters by **27.6%** ($35.8\text{M} \to 25.9\text{M}$) while advanc-

(a) Ground Truth (GT) **(b) ESOD (Baseline)** **(c) HESOD (Ours)**

Fig. 4. Qualitative comparison on SeaPerson (`rgb_x0080/google_P000_115344`, *Surfing*): ESOD misses weak micro targets from feature degradation, while HESOD reliably recalls them with accurate localization.

Table 4. Module ablation on SeaPerson (2048 × 2048).

Method / Config	mAP @.5	mAP @.5:.95	BPR	GFLOPs	FPS
(1) Baseline (BCE)	0.750	0.320	0.947	**202.4**	**85.7**
(2) Semantic-only*	0.769	0.325	0.991	266.8	73.5
(3) Spectral-only*	0.770	0.327	**0.992**	267.4	71.7
(4) Spectral-Pooled*	0.767	0.324	0.988	263.4	72.7
(5) Dual-Concat	**0.772**	0.326	0.986	281.2	69.8
(6) Dual+SABL	0.771	0.323	0.990	263.7	73.4
(7) Dual+ISPP	0.771	**0.328**	0.988	217.6	77.3
(8) HESOD (Full)	0.769	0.323	0.990	209.1	77.1

*Trained with soft-coverage loss $\mathcal{L}_{\mathrm{cover}}$. **Bold**: best; underline: second-best.

Table 5. Size-bucket recall on UAVDT (1280 × 1280).

Method / Config	V-Tiny ($< 16^2$)	Tiny (16^2–32^2)	Small (32^2–96^2)	Med/L ($> 96^2$)	Total (All)
GT Count	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(1) Baseline (BCE)	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(2) Semantic-only*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(3) Spectral-only*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(4) Spectral-Pooled*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(5) Dual-Concat	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(6) Dual+SABL	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(7) Dual+ISPP	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
(8) HESOD (Full)	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*

*Trained with $\mathcal{L}_{\mathrm{cover}}$. Bins in px^2. **Bold**: best.

ing $\mathrm{mAP}@.5:.95$ to a benchmark peak of **0.328** at 77.3 FPS. Integrating scale-aware Wasserstein regression in Arm (8) strategically balances coarse bounding mAP with extreme sub-16px localization, pushing *Very Tiny* recall to its project-wide maximum of **77.19%** (+2,605 recovered micro-targets over baseline) and minimizing GFLOPs to **209.1** via tighter patch clustering. HESOD Full thereby realizes the optimal Pareto frontier across accuracy, micro-recall, and edge efficiency.

Scale-Stratified Gains and Error Bottleneck Shift. Detailed breakdown across ground-truth instances (Table 5 and Table 6) reveals that target recovery is strongest on the dominant *Tiny* stratum (+4.79 pp recall, recovering 8,380 targets) and the challenging *Very Tiny* stratum (+3.16 pp recall). Error diagnosis confirms that dual-evidence routing and coverage optimization curtail selector-dropped error from 25.6% in baseline ESOD to 19.2%, effectively curing early-stage false-negative pruning and shifting the remaining detection challenge to downstream refinement.

Table 6. Size-bucket recall on SeaPerson (2048 × 2048).

Method / Config	V-Tiny ($< 16^2$)	Tiny (16^2–32^2)	Small (32^2–96^2)	Med/L ($> 96^2$)	Total (All)
GT Count	*82,417*	*174,816*	*42,987*	*155*	*300,375*
(1) Baseline (BCE)	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only*	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only*	75.23%	**91.69%**	**95.89%**	86.45%	87.77%
(4) Spectral-Pooled*	76.13%	91.36%	95.50%	**87.10%**	87.77%
(5) Dual-Concat	76.00%	91.50%	95.68%	80.00%	87.84%
(6) Dual+SABL	76.67%	91.49%	94.77%	80.65%	**87.89%**
(7) Dual+ISPP	75.86%	91.25%	95.85%	**87.10%**	87.68%
(8) HESOD (Full)	**77.19%**	90.70%	94.89%	83.23%	87.59%

*Trained with $\mathcal{L}_{\mathrm{cover}}$. Bins in px^2. **Bold**: best; underline: second-best.

Qualitative Comparison. As illustrated in Fig. 4, baseline ESOD suffers from false-negative drops on weak-contrast micro persons, whereas HESOD's dual-evidence spatial routing and scale-aware regression accurately localize all tiny instances without extraneous false alarms.

5. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence selection framework tailored for recall-safe, high-resolution small object detection in aerial and maritime vision. By addressing the critical vulnerability of semantic feature collapse in single-evidence selective computation, HESOD introduces a dual-evidence priority head that combines semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width). In addition, we formulated an object-level soft-coverage loss that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, drastically reducing early false-negative dropping of tiny targets. Paired with a lightweight ISPP-based decoupled detection head and scale-aware bounding-box regression (SABL), HESOD achieves substantial improvements in both detection accuracy and tiny-object recall across challenging benchmarks including UAVDT and SeaPerson, while maintaining real-time inference speeds. Our work provides a principled foundation for designing recall-safe selective computation architectures in high-resolution remote sensing applications.

6. REFERENCES

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